

ToxiTaARGET: a multi-omics database for toxicant-responsive molecular targets

Ravindra Kumar^{1,2,*}, Tianyi Fu^{1,2,*}, Prashant Kumar Kuntala^{3,4}, Benpeng Miao^{3,4}, Shuhua Fu^{1,2}, Daofeng Li^{3,4}, TaARGET II Consortium, Frederick L. Tyson⁵, Marisa S. Bartolomei⁶, Cheryl L. Walker^{7,8}, Ting Wang^{3,4,9}, and Bo A. Zhang^{1,2,#}

1. Department of Developmental Biology, Washington University School of Medicine, St. Louis, MO 63110, USA
2. Center of Regenerative Medicine, Washington University School of Medicine, St. Louis, MO 63110, USA
3. Department of Genetics, Washington University School of Medicine, St. Louis, MO 63110, USA
4. Edison Family Center for Genome Sciences and Systems Biology, Washington University School of Medicine, St. Louis, MO 63110, USA
5. National Institute of Environmental Health Sciences, NIH, 111 TW Alexander Dr., Durham, NC 27709, USA.
6. Department of Cell and Developmental Biology, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA 19104, USA
7. Dan L. Duncan Comprehensive Cancer Center, Baylor College of Medicine, Houston, TX 77030, USA
8. Center for Precision Environmental Health, Baylor College of Medicine, Houston, TX 77030, USA
9. McDonnell Genome Institute, Washington University School of Medicine, St. Louis, MO 63110, USA

* Equal contribution

To whom correspondence should be addressed

Bo A. Zhang, Ph.D.

Email: bzhang29@wustl.edu

Supplementary Materials

- **TaARGET II consortium (alphabetical)**
- **Supplementary Methods**

- **TaARGET II consortium (alphabetical)**

Integrative analysis leads and coordination

Benpeng Miao, Ting Wang, Bo A. Zhang,

Integrative analysis

Cristian Coarfa, Justin A. Colacino, Shuhua Fu, Sandra L. Grimm, Yan Jin, Ravindra Kumar, Prashant Kumar Kuntala, Bongsoo Park, Wanqing Shao, Laurie K. Svoboda

Integrative data production and processing

Gregory E. Crawford, Robert B. Hamanaka, Claudia Lalancette, Daofeng Li, Shaopeng Liu, Benpeng Miao, Heather B. Patisaul, Maureen A. Sartor, Tim Wiltshire, Xiaoyun Xing, Bo A. Zhang

TaARGET II Data Production and Processing contributor

Nicole E. Allard, Raymond G. Cavalcante, Rengul Cetin-Atalay, Yujie Chen, Youngshim Choi, Alan Du, Elisa Ruiz-Echartea, Jackson P. Fredenburg, Tianyi Fu, Jaclyn M. Goodrich, Silas Hsu, Brian M. Horman, Yiran Hou, Rahul Jangid, Tamara R. Jones, Tiffany A. Katz, SunHong Kim, Prashant Kumar Kuntala, Yemin Lan, Bethany Latham, Tandao Li, Yan Li, Sydney Lierz, Siyu Liu, Juheon Maeng, Angelo Y. Meliton, Rachel K. Morgan, Kari Neier, Jackson P. Parker, Bambarendage P.U. Perera, Jayant M. Pinto, Deepak Purushotham, Dhivyaa Rajasundaram, Palanivel Rengasamy, Christine A. Rygiel, Alexias Safi, Erica Pehrsson, Kaitlyn A. Sun, Vinesh Vinayachandran, Kai Wang, Parker S. Woods, Bonnie HY Yeung, Jinhu Yin, Yu Zhang, Xiaoyu Zhuo

Co-principal investigators

Cristian Coarfa, Gregory E. Crawford, Heather Lawson, Michael Province

Scientific program management

Lisa H. Chadwick, Christopher G. Duncan, Kimberly A. McAllister, Frederick L. Tyson

Principal Investigators

David Aylor, Marisa S. Bartolomei, Shyam Biswal, Dana C. Dolinoy, Gökhan M. Mutlu, Sanjay Rajagopalan, Wan-Yee Tang, Cheryl Lyn Walker, Ting Wang

Affiliations

Department of Developmental Biology.

Center of Regenerative Medicine.

Washington University School of Medicine, St. Louis, MO 63108, USA

Shuhua Fu, Kumar Ravindra, Tiandao Li, Shaopeng Liu, Tianyi Fu, Yan Jin, Bo A. Zhang

Department of Genetics.

Washington University School of Medicine, St. Louis, MO 63108, USA

Benpeng Miao, Daofeng Li, Xiaoyun Xing, Wanqing Shao, Erica Pehrsson, Prashant Kumar Kuntala, Deepak Purushotham, Xiaoyu Zhuo, Silas Hsu, Ju Heon Maeng, Yujie Chen, Alan Du, Yiran Hou, Heather Lawson, Michael Province, Ting Wang

Dan L. Duncan Comprehensive Cancer Center.

Center for Precision Environmental Health.

Baylor College of Medicine, Houston, TX 77030, USA

Tiffany A. Katz, Sandra Grimm, Rahul Jangid, Cristian Coarfa, Cheryl Walker

Department of Environmental Health and Engineering.

School of Public Health, Johns Hopkins University, Baltimore, MD 21205, USA

Bongsoo Park, Bonnie HY Yeung, Dhivyaa Rajasundaram, Youngshim Choi, Palanivel Rengasamy, Vinesh Vinayachandran, SunHong Kim, Jinhu Yin, Shyam Biswal,

Department of Environmental Health Sciences.

Department of Nutritional Sciences.

Department of Pharmacology.

Department of Computational Medicine and Bioinformatics.

Epigenomics Core.

University of Michigan School of Medicine, Ann Arbor, MI 48109 USA

Raymond G. Cavalcante, Justin A. Colacino, Jaclyn M. Goodrich, Tamara R. Jones, Claudia Lalancette, Siyu Liu, Rachel K. Morgan, Kari Neier, Bambarendage P.U. Perera, Christine A. Rygiel, Maureen A. Sartor, Laurie K. Svoboda, Kai Wang, Dana C. Dolinoy,

Department of Medicine.

Section of Pulmonary and Critical Care Medicine.

The University of Chicago, Chicago, IL 60637, USA

Robert B. Hamanaka, Angelo Y. Meliton, Jayant M. Pinto, Parker S. Woods, Kaitlyn A. Sun, Rengul Cetin-Atalay, Yan Li, Gökhan M. Mutlu.

Department of Cell and Developmental Biology.

Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA 19104, USA

Christopher Krapp, Yemin Lan, Marisa S. Bartolomei.

Division of Medical Genetics.

Department of Pediatrics.

Duke University School of Medicine, Durham, NC 27710, USA

Gregory E. Crawford

Department of Biological Sciences,

North Carolina State University, Raleigh, NC 27695

Heather B. Patisaul, Nicole E. Allard, Thomas I. Konneker, Alexias Safi, Jacob D. Fredenburg, Jackson P. Parker, Sydney Lierz, Brian M. Horman, Bethany Latham, David L. Aylor

Eshelman School of Pharmacy

University of North Carolina at Chapel Hill, Chapel Hill, NC

Tim Wiltshire

Cardiovascular Research Institute, School of Medicine, Case Western Reserve University, OH 44106, USA

Sanjay Rajagopalan

Department of Environmental and Occupational Health.

University of Pittsburgh School of Public Health, Pittsburgh, PA 15216, USA

Wan-yee Tang

• **Supplementary Methods**

Animal exposure paradigm and tissue collection

All experiments used C57BL/6 (B6; Jackson Laboratory, Bar Harbor, ME) mice, except for the lead (Pb) and phthalate (DEHP) studies, which employed wild-type non-agouti (a/a) mice from a >230-generation colony of viable yellow agouti (Avy) mice. These a/a mice are genetically uniform and share 93% genetic similarity with the C57BL/6J strain. Mice were exposed to toxicants perinatally through maternal diet, drinking water, or air, from pre-conception until weaning. Specifically, two weeks before mating, virgin female dams (6–8 weeks old) were randomly assigned to one of nine exposure groups or a control group.

- 1) 10g/kg bw/day diet of BPA (BPA10mg);
- 2) 10µg/kg bw/day diet of BPA (BPA10µg);
- 3) 25 mg/kg diet of DEHP (DEHP);
- 4) 32 parts per million (ppm) Pb-acetate drinking water (Pb);
- 5) PM <2.5 micrometer generated by Johns Hopkins University (PM2.5-JHU);
- 6) PM <2.5 micrometer generated by University of Chicago (PM2.5-CHI);
- 7) 0.5 mg/kg BW tributyltin drinking water (TBT);
- 8) 1 ug/kg per BW of dioxin via oral gavage (TCDD);
- 9) 10 parts per billion (ppb) sodium arsenite drinking water (As).

Female mice (dams) were paired with virgin males (7–9 weeks old) after two weeks of exposure to the specified experimental diets or water. This exposure continued throughout pregnancy and lactation until the offspring were weaned at postnatal day 21 (PND21). The animals were kept in polycarbonate-free cages under a 12-hour light/dark cycle. All procedures received approval from the Institutional Animal Care and Use Committees (IACUC) at each participating site and followed the guidelines set by the NIEHS TaRGET II Consortium, ensuring the highest standards of animal welfare.

Pb, TBT, and TCDD exposures were delivered through different methods: Pb and TBT via drinking water freely available to animals, BPA (both high and low doses) and DEHP through diet, and TCDD via oral gavage. Additionally, PM2.5 exposures were administered using a breathing control chamber as scheduled.

As exposure

All animal procedures received approval from the University of Pittsburgh Institutional Animal Care and Use Committee. Mice were fed an AIN-93G diet (Research Diets, NJ) that was free of fishmeal and rice products to prevent arsenic contamination. Their drinking water contained either 0 or 10 parts per billion (10 ppb, equivalent to 10µg/L/g.o.d) sodium arsenite ($\geq 90\%$ purity; Sigma-Aldrich), renewed every other day. Both the diet and water were tested for arsenic

content batch by batch. Female nulliparous mice were exposed to either 0 or 10 ppb inorganic arsenic (iAs) in their drinking water starting two weeks before mating and continued until the end of lactation, totaling eight weeks. After weaning, the F1 offspring were no longer exposed to iAs.

BPA exposure

For BPA exposure, a modified AIN-93G diet supplemented with 50 mg/kg BPA (TD.06156; Harlan Teklad) was used for the high-dose group, and a modified AIN-93G diet supplemented with 50 µg/kg BPA (TD.110337; Harlan Teklad) was used for the low-dose group, as previously described¹. All diet ingredients were sourced from Harlan Laboratories (Madison, WI), with BPA obtained from Sigma-Aldrich (St. Louis, MO). Drinking water was provided in polypropylene bottles. The estimated daily BPA intake was approximately 10 mg/kg body weight (bw) for the high-dose group and 10 µg/kg bw for the low-dose group. Throughout the experiment, animals were maintained on a phytoestrogen-free modified AIN-93G chow (Envigo TD.95092; 7% corn oil diet, Harlan Teklad).

DEHP exposure

DEHP was dissolved in corn oil (Envigo) to prepare a stock solution used to generate a customized 7% corn oil chow for experimental exposures. The DEHP concentration was selected to achieve a target maternal dose of 5 mg/kg/day, based on an estimated body weight of 25 g and a daily chow intake of 5 g for pregnant and nursing female mice. This dosing strategy was informed by previous studies demonstrating obesity-related phenotypes following perinatal DEHP exposure at similar levels², and falls within the range of human exposure levels³. Throughout the experiment, animals were maintained on phytoestrogen-free modified AIN-93G chow (Envigo Td.95092, 7% corn oil diet, Harlan Teklad).

Pb exposure

Pb-acetate drinking water was prepared by dissolving Pb(II) acetate trihydrate (Sigma-Aldrich) in distilled water to a final concentration of 32 ppm, modeling human-relevant perinatal exposure levels. Previous studies confirmed that this dosing regimen results in maternal blood lead levels (BLLs) ranging from 16 to 60 µg/dL (mean: 32.1 µg/dL)⁴. All exposure water was prepared in a single batch, and Pb concentrations were validated using inductively coupled plasma mass spectrometry (ICP-MS; NSF International; limit of detection = 1.0 µg/L). Animals were maintained on phytoestrogen-free modified AIN-93G chow (Envigo Td.95092, 7% corn oil diet, Harlan Teklad) for the duration of the experiment.

TBT exposure

For TBT exposure, animals received tributyltin (TBT) chloride (96% purity; Sigma-Aldrich) in their drinking water, following established protocols. Water intake was monitored in non-pregnant females to estimate the TBT dose. Notably, water consumption rose sharply for about three days after parturition, coinciding with lactation onset. With an average intake of 10 mL/day, animals ingested 200 mL of 3.07 µM TBT solution weekly, equating to roughly 0.5 mg/kg body weight/day. This dosage is below the lowest threshold of the developmental NOAEL (5.8–20 mg/kg body weight) outlined in the Concise International Chemical Assessment Documents (CICADs) for developmental toxicity (Benson, 1999). Throughout the study, animals were fed a phytoestrogen-free modified AIN-93G diet (Envigo Td.95092, Harlan Teklad) containing 7% corn oil.

TCDD exposure

For TCDD exposure, female mice were given 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD; Cambridge Isotopes, cat# ED-901-C) dissolved in corn oil through oral gavage. The dams received three doses of 0.34 µg/kg body weight each, totaling 1 µg/kg during the entire exposure period from preconception to weaning. Dosing took place at three time points: two weeks before mating, around embryonic day 7, and approximately postnatal day 14. Control animals were given pure corn oil. All animals were kept on phytoestrogen-free modified AIN-93G chow (Envigo Td.95092, 7% corn oil diet, Harlan Teklad) throughout the study.

PM2.5-JHU exposure

Animals were subjected to either PM2.5 (150 µg/m³) or filtered air (FA) in dedicated exposure chambers. Exposure took place five days a week (Monday to Friday) from 9:00 AM to 5:00 PM, totaling 8 hours daily over seven weeks. Before starting, all animals were acclimated in the housing facility for one week. During the exposure period, each group (FA or PM2.5) was alternated between two different chambers to reduce chamber-specific effects. No significant differences in locomotor activity or mobility were observed in the mice before or during the exposure, indicating consistent baseline physical activity across groups. Throughout the study, mice had access to control water and food ad libitum during the PM2.5 exposure.

PM2.5-CHI exposure

Dams were placed in either the PM2.5 (135 µg/m³) or control chamber for 8 hours daily and mated overnight with males until pregnancy was confirmed, as previously described⁵. Once pregnant, they continued exposure to their respective conditions (PM2.5 or filtered air) for 8 hours per day until birth. After that, both dams and pups experienced the same exposure (PM2.5 or filtered air) for 8 hours daily until the pups were 3 weeks old. The PM2.5 was concentrated from ambient air in Chicago using the Versatile Aerosol Concentration Enrichment System (VACES), as described earlier⁵. The chemical makeup of this PM2.5 has been reported before⁵. Control mice were exposed to filtered air in an identical VACES chamber, with a Teflon filter on the inlet to remove all particles. Throughout the study, mice had access to water and food ad libitum.

Tissue collection

By postnatal day 21 (PND21), all offspring were weaned and switched to control water and chow. Across the consortium, one set of male and female mice from each exposure group was euthanized at 3 weeks for tissue collection and molecular analysis. A second set of both sexes was kept until 5 months, and a third set until 10 months. The study assesses animals from all three time points: 3 weeks, 5 months, and 10 months.

This study involved several exposure groups: 25 As, 34 high BPA, 33 low BPA, 32 DEHP, 35 Pb, 51 TBT, 24 TCDD, 24 PM2.5-JHU, 46 PM2.5-CHI, and 227 control mice, totaling 531 animals (n = 531). Blood, liver, cortex, heart, lung, and other tissues were collected from all subjects, with no exclusions. Mice were fasted for four hours during the light cycle in the morning for organ and tissue collection. In the afternoon, they were euthanized with CO₂, followed by cardiac puncture and whole-body saline perfusion (Sigma). The cortex and liver were dissected, flash frozen in liquid nitrogen, and stored at -80°C, following protocols from the

TaRGET II Methods Working Group. Blood was drawn into EDTA tubes (Fisher, Cat. #365974), and plasma was separated via centrifugation. Red blood cells were lysed using erythrocyte lysis buffer (Qiagen, Cat. #79217), then the white blood cells were washed in PBS and resuspended in Buffer RLT (Qiagen, Cat. #79216) for DNA extraction. SOPs for tissue collection are available at <https://targetepigenomics.org/documents/>, including SOP_Blood_Processing, SOP_Mouse_Liver_Collection, and SOP_Whole_Mouse_Perfusion. One investigator administered the treatment to each mouse and was aware of the group assignment, while all other investigators performing molecular assays were blinded to the treatment groups until bioinformatics analysis.

DNA extraction and whole-genome bisulfite sequencing (WGBS)

DNA extraction was carried out using the AllPrep DNA/RNA/miRNA Universal Kit (Qiagen, Cat. #80224). Genomic DNA (gDNA) was used to prepare WGBS libraries at the University of Michigan Epigenomics Core. gDNA was quantified with the Qubit BR dsDNA kit (Fisher, Cat. #Q32850) and quality-checked using Agilent's Genomic DNA TapeStation Kit (Agilent, Cat. #A63880). One WGBS library was made for each animal, each labeled with a unique dual index for deconvolution. For each sample, 200 ng of gDNA was spiked with 0.5% unmethylated lambda DNA (Cat. no. D1521; Promega) and sheared with a Covaris S220 (10% Duty Factor, 140W Peak Incident Power, 200 Cycles/Burst, 55s). A 2 µl aliquot of sheared gDNA was used to assess shearing quality with an Agilent High Sensitivity D1000 Kit (Agilent, Cat. #G2991AA). After shearing, the remaining gDNA was concentrated using a Qiagen PCR Purification column and processed for end-repair and A-tailing. Ligation of methylated adapters was performed overnight at 16°C. The ligation products were purified with AMPure XP Beads (Fisher, Cat. #NC9933872), then subjected to bisulfite conversion using the Zymo EZ DNA Methylation Kit (Zymo, Cat. #D5001), following the protocol: 55 cycles of 95°C for 30 seconds, then 55°C for 15 minutes. Post-conversion, final libraries were amplified over 10 PCR cycles with KAPA Uracil+ Ready Mix (Fisher, Cat. #501965287) and NEB dual indexing primers. Libraries were cleaned with AMPure XP beads, their concentration measured with the Qubit BR dsDNA Kit, and their size distribution checked on the Agilent High Sensitivity D1000 TapeStation Kit. Before pooling, each library was quantified using the KAPA Library Quantification Kit (Fisher, Cat. #501965234). Two pools of 18 libraries each were created, mixing individuals from different litters, sexes, and exposures to reduce batch effects. These pools were sequenced on an Illumina NovaSeq6000 S4 200 cycle flow cell (PE-100) at the University of Michigan Advanced Genomics Core. All enzymes used were purchased from New England Biolabs unless stated otherwise. Adapters with methylated cytosines were synthesized by Integrated DNA Technologies (IDT).

RNA extraction and RNA-seq library construction

Total RNA was extracted using TRIzol Reagent (Thermo Fisher Scientific, 15596026), Phasemaker Tubes (Thermo Fisher Scientific, A33248), and RNA Clean & Concentrator-5 (Zymo Research, R1013). Briefly, rat brain tissues were homogenized in 1 ml of TRIzol Reagent, and the lysates were transferred to a pre-spined Phasemaker Tube. Then, 0.2 ml of chloroform was added, and the tubes were shaken for 15 seconds. The mixture was centrifuged at 16,000 g for 5 minutes, and the aqueous phase was collected into a microcentrifuge tube. An equal volume of ethanol was added to the aqueous phase. RNA purification, including DNase treatment, was performed according to the RNA Clean & Concentrator kit manual. Ribosomal RNAs were removed from 500 ng of total RNA using the NEBNext rRNA Depletion kit (NEB, E6310). Bypassing mRNA isolation, RNA-seq libraries were constructed from 10 ng of rRNA-depleted total RNA using the Universal Plus mRNA-seq kit (TECAN, 0520-A01) following the kit instructions. Paired-end sequencing was conducted with 2×75 bp, 2×100 bp, and 2×150 bp reads on Illumina platforms.

ATAC-seq library construction

ATAC-seq was generated using the omni ATAC-seq protocol for frozen tissues^{6,7}. In brief, mouse tissues were homogenized in 2 ml of cold 1× homogenization buffer. Nuclei were layered from the tissue lysate with iodixanol solution. 50,000 nuclei were used in the transposition reaction with 100 nM of Transposase (Illumina, 20034197). The ATAC-seq libraries were prepared by amplifying for 9 cycles on a PCR machine with NEBNext High-Fidelity 2× PCR master mix (NEB, M0541). 2 × 75bp paired-end sequencing was run for all libraries on the Illumina platforms.

Raw sequence data and processing

Omics datasets were collected from mice at ages 3, 20, and 40 weeks that were exposed to pollutants during early development. These pollutants included arsenic (As), lead (Pb), bisphenol A (BPA), tributyltin (TBT), di-2-ethylhexyl phthalate (DEHP), 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD), and particulate matter smaller than 2.5 µm (PM_{2.5}). The datasets comprised 1,043 RNA-seq samples, 638 chromatin accessibility profiles, 659 histone modification profiles, 529 DNA methylation data (458 WGBS and 71 RRBS) and 744 Infinium Mouse Methylation BeadChip entries.

RNA-seq data processing followed methods from previous studies. Raw FASTQ files underwent trimming with Cutadapt⁵ (v1.16; --quality-cutoff=15,10 --minimum-length=36), quality control with FastQC (v0.11.7), and mapping to the mouse genome (mm10) using STAR⁵ (v2.5.4b; --quantMode TranscriptomeSAM --outWigType bedGraph --outWigNorm RPM). These steps were performed with our custom pipeline, TaRGET-II-RNA-seq-pipeline (<https://github.com/Zhang-lab/TaRGET-II-RNAseq-pipeline>). Subsequently, gene expression levels in normal and exposed samples were quantified using featureCounts⁵ (v1.5.1) within the same pipeline, based on GENCODE vM20 gene annotation for the mouse genome.

Raw fastq of ATAC-seq data were processed by our own built pipeline based on AIAP⁸, the TaRGET-II-ATAC-seq-pipeline (<https://github.com/Zhang-lab/TaRGET-II-ATACseq-pipeline>) that integrated AIAP packages containing optimized QC reports and analysis pipeline with default parameters to generate the open chromatin regions (OCRs). Then, the consensus regions of OCRs across all exposure and control samples were generated with Index (<https://github.com/Altius/Index>) method used for the downstream analysis. The ATAC-seq signals of consensus OCRs were calculated by using the intersectBed method.

The TaRGET-II-WGBS-pipeline (https://github.com/Zhang-lab/WGBS_analysis) was built by using the Cutadapt⁹ (v1.16; --quality-cutoff=15,10 --minimum-length=36) and Bismark¹⁰ (v0.19.0; --bowtie2 -X 1000 --score_min L,0,-0.6 -N 0 --multicore 2 -p 4) to do the trimming and mouse genome mapping. The pipeline also incorporated quality control, generated user-friendly files for computational analysis, and output genome browser tracks for data visualization.

Normalization and Batch Effect Correction for RNA-seq and ATAC-seq Data

RNA-seq and ATAC-seq datasets were normalized to correct for batch effects arising from different data production centers. Samples with the proportion of reads aligned to chrY relative to reads aligned to chrX were used to verify the sex information from metadata. In general,

RNA-seq and ATAC-seq data generated within the same production center were analyzed together. Gene counts from RNA-seq were first normalized using the relative log expression (RLE) method implemented in *edgeR*. Subsequently, factor analysis was performed across exposure groups by applying the residuals calculation of the *RUVr* function in *RUVSeq*¹¹. To further remove unwanted technical variations (k=3) unrelated to exposures, including biases from tissue dissection and library preparation, a general linear model fitting process was employed as in previous studies^{12,13}.

To eliminate bias across production centers, the dataset of exposures was first matched to corresponding control samples and then corrected using a modified batch-free data correction method inspired by the ComBat-seq model¹⁴. Typically, a theoretical “batch-free” control dataset for each sex, called Ctrl_P (CP), was created. This represents the estimated empirical distribution of all control samples sharing the same variation factors (tissue, age, sex) across different production centers. For each sex, the pseudo-control signal is defined as the average signal across all control samples of that sex from all centers.

$$\overline{X}_{CP} = \frac{1}{N_{CC}} \sum_{i=1}^{N_{CC}} X_{i,CC}$$

Since the Euclidean distance between each production center's control (CC) and CP indicates the systematic batch effect (see **Figure: pseudo-control correction**), we use this distance as a normalization weight factor (*nw*). This correction adjusts gene expression levels for both control and exposed samples. The expression of the center control (CC) for each production center is defined as

$$Exp_{CC_i} = \sum_m (Exp_{i_j}) / m$$

Exp_{i_j} is the expression of gene *i* in sample *j* out of *m* control samples, and normalization weight factor (*nw*) is defined as the vector from the center control (CC) to pseudo-control Ctrl_P (CP), and the normalized expression of gene *i* ($N_Exp_{ESj_i}$) of exposed sample *j* (ES*j*) are:

$$\overrightarrow{nw} = \overrightarrow{CC,CP} = \sqrt{\sum_n (Exp_{CC_i} - Exp_{CP_i})^2}$$

$$N_Exp_{ESj_i} = \overrightarrow{nw} \cdot Exp_{ESj_i}$$

Pseudo-controls for females and males were created by averaging the signal from all control samples of each sex across various production centers. Each control sample was then aligned with its corresponding pseudo-control by scaling its mean signal to match the pseudo-control reference, done separately for each sex. To preserve biological differences between control and exposed samples, all samples from each center were adjusted based on the fold changes identified during the alignment process.

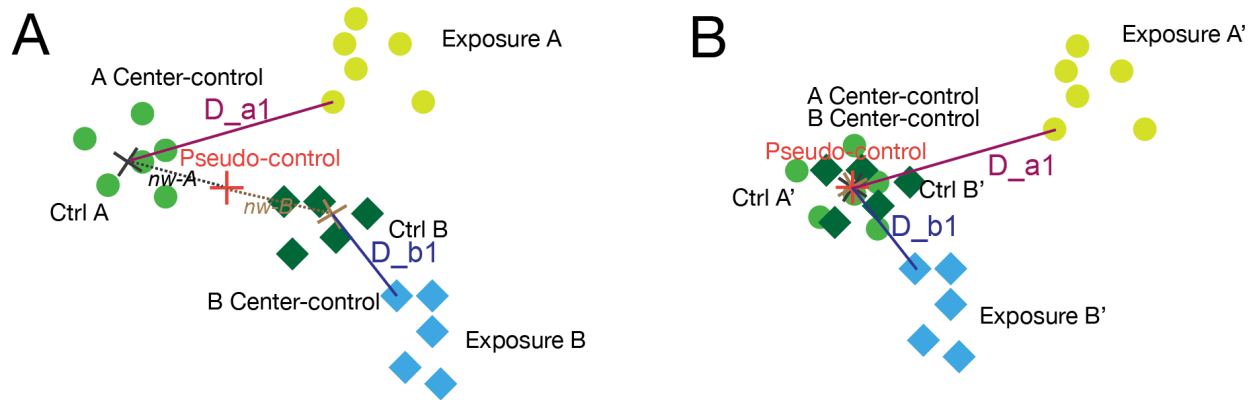


Figure: pseudo-control correction. A) Data distribution of control samples and exposed samples from two cancer data sources in a two-dimensional space. A_Center-control is the centroid of all controls from center A, and B_Center-control is the centroid of all controls from center B. D_a1 represents the vector from exposed sample a1 to A_Center-control, while D_b1 is from exposed sample b1 to B_Center-control. nw-A is the normalization factor for center A samples, and nw-B is for center B samples. B) Center A samples were normalized using nw-A, with the D_{a1} remaining unchanged. Similarly, center B samples were normalized with nw-B, and the D_{b1} has remained.

Read counts of ATAC peaks across all data production centers were normalized in the same way as the RNA-seq data normalization above.

Differentially analysis

Differentially expressed genes (DEGs) between control and exposure samples under varying conditions were identified using DESeq2¹⁵ (v1.34.0) as previous studies¹⁶⁻¹⁸. Consortium-normalized read count data were used to identify the DEGs between the exposed and control samples. Genes were considered significantly differentially expressed if they met the following stringent criteria: an adjusted p-value (Benjamini-Hochberg correction) less than 0.001 and an absolute log2 fold change greater than log2(1.5), equivalent to a 1.5-fold change in expression.

Differentially accessible regions (DARs) between control and exposure samples were identified using edgeR¹⁹ (v3.36.0), as previous study^{8,20,21}. Consortium-normalized ATAC-seq reads were used to identify DARs, with significance determined by the following cutoffs: an absolute log2 fold change greater than log2(1.5) (a 1.5-fold change in accessibility) and a false discovery rate (FDR) less than 0.01. These criteria balance sensitivity and specificity, ensuring robust detection of chromatin accessibility changes associated with exposure.

Differential methylation regions (DMRs) between control and exposure samples were identified using a method adapted from previous studies²²⁻²⁵. Methylation levels were quantified using whole-genome bisulfite sequencing (WGBS) data. Briefly, raw WGBS reads were aligned to the reference genome using Bismark¹⁰ (v0.23.1) with default parameters, followed by deduplication to remove PCR artifacts. Methylation levels at individual CpG sites were calculated as the ratio of methylated reads to total reads (methylated + unmethylated) at each site, with a minimum coverage of 10 reads per site to ensure reliability. Samples with fewer than 15 million CpG sites or those with less than 60% of CpG sites having a methylation level below 0.7 are excluded from the analyses. Samples with the proportion of reads aligned to chrY relative to reads aligned to chrX greater than 4% were considered male, while those with proportions less than 4% were considered female. If a sample's deduced sex information conflicts with the labeled

sex information, it will be excluded from subsequent analyses. For each sample, the mean methylation values for each 1k window across the mm10 genome were calculated and processed using the *normalize.quantiles* function in the preprocessCore R package (v1.60.2). The normalized data were then used in PCA analysis.

DMRs were identified through a genome sliding approach, as previous studies²⁴⁻²⁶: the genome was divided into sliding 200bp windows, the methylated and unmethylated reads counts in each window were calculated in both control samples and exposed samples, and then a Chi-squared test was applied to identify DMRs. Regions were defined as differentially methylated if they exhibited a methylation level difference of at least 0.1 (a 10% absolute difference in methylation between control and exposed condition) and a Q-value (adjusted for multiple testing using the BH method) less than 0.1. Additionally, only regions with at least 2 CpG sites and a minimum average coverage of 20 reads per site were considered to enhance statistical power and reduce noise in the analysis, and the neighboring regions were merged to maximize the width of DMRs.

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